



NQT 2025

Top 30 Most Asked JAVA Interview Q&A 2025 Prime, Digital, Ninja Role

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TCS NQT 2025: Top 30 Java Interview Questions with Answers

1. What are the features of Java?

Java is an object-oriented, high-level programming language known for its platform independence, achieved through the Java Virtual Machine (JVM). It supports multithreading, automatic memory management (garbage collection), and is widely used for building cross-platform applications. Java is secure, robust, and follows the principle of Write Once, Run Anywhere (WORA).

2. What is JVM, JRE, and JDK?

JVM (Java Virtual Machine) executes Java bytecode on any platform. JRE (Java Runtime Environment) includes the JVM and standard libraries necessary to run Java programs. JDK (Java Development Kit) contains JRE and development tools like compilers and debuggers to develop Java applications.

3. What is the difference between '==' and '.equals()' in Java?

'==' is used to compare primitive data types or reference memory locations for objects. '.equals()' is a method used to compare the actual content of two objects, typically overridden in classes like String to perform value-based comparisons.

4. What is a Constructor in Java?

A constructor is a special method used to initialize objects. It has the same name as the class and does not have a return type. Constructors can be default (no parameters) or parameterized. They are called automatically when an object is created.

5. Explain OOPs concepts in Java.

Java is based on Object-Oriented Programming (OOP) principles: Inheritance (reuse existing code), Encapsulation (binding data and methods), Abstraction (hide internal implementation), and Polymorphism (one interface, multiple implementations). These principles make code modular, reusable, and easier to maintain.

6. What is the difference between Overloading and Overriding?

Method Overloading occurs when multiple methods have the same name but different parameters in the



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same class. Method Overriding happens when a subclass provides a specific implementation of a method declared in its parent class with the same signature.

7. What is the use of the 'final' keyword?

The 'final' keyword in Java is used to define constants, prevent method overriding, and prevent inheritance. A final variable cannot be changed, a final method cannot be overridden, and a final class cannot be extended.

8. What is the difference between ArrayList and LinkedList?

ArrayList uses a dynamic array to store elements, offering fast random access but slower insertions/deletions. LinkedList uses nodes and is faster for insertions/deletions but slower for random access. Both implement the List interface.

9. What is the difference between abstract class and interface?

An abstract class can have both abstract and concrete methods, constructors, and member variables. Interfaces can only contain abstract methods (until Java 7), though Java 8+ allows default and static methods. Interfaces support multiple inheritance, while abstract classes do not.

10. What is Exception Handling in Java?

Exception handling in Java is done using try, catch, finally, throw, and throws. It helps manage runtime errors, ensuring the program does not terminate unexpectedly. Exceptions are objects that describe an error condition.

11. What is the difference between throw and throws?

'throw' is used to explicitly throw an exception in code. 'throws' is used in method declarations to indicate what exceptions might be thrown by the method.

12. What is multithreading in Java?

Multithreading allows concurrent execution of two or more threads. Threads share the same memory but run independently. Java provides the Thread class and Runnable interface to create and manage threads.



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13. What is the use of the 'this' keyword?

'this' refers to the current instance of the class. It is used to distinguish between local variables and instance variables, and to call other constructors in the same class.

14. What are access specifiers in Java?

Access specifiers control the visibility of classes, methods, and variables. 'private' is accessible only within the class, 'default' (no modifier) within the package, 'protected' within the package and subclasses, and 'public' is accessible everywhere.

15. What is garbage collection in Java?

Garbage collection is a process by which the JVM automatically deletes unused objects from memory to free up resources. This helps prevent memory leaks and enhances performance.

16. What are Wrapper Classes in Java?

Wrapper classes convert primitive types (int, char, etc.) into objects. They are useful when working with collections or when object-based manipulation is needed. Examples include Integer, Character, Double, etc.

17. What is the difference between String, StringBuilder, and StringBuffer?

String is immutable, meaning any change results in a new object. StringBuilder is mutable and faster than StringBuffer, but it is not thread-safe. StringBuffer is also mutable but is synchronized, making it thread-safe and suitable for multi-threaded environments.

18. What is a static variable and method?

Static variables and methods belong to the class rather than any object instance. They are shared among all instances. A static method can be called without creating an object and cannot access non-static members directly.

19. Can a class inherit multiple classes in Java?



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No, Java does not support multiple inheritance through classes to avoid ambiguity and complexity. However, it supports multiple inheritance via interfaces, allowing a class to implement multiple interfaces.

20. What is the difference between Stack and Heap memory in Java?

Stack memory stores method call frames, local variables, and references. Heap memory is used to store actual objects and class instances. Stack memory is faster and automatically managed, whereas heap memory is managed by the garbage collector.

21. What is the difference between 'continue' and 'break' in Java?

'break' exits the loop immediately, whereas 'continue' skips the current iteration and proceeds with the next iteration of the loop. Both are used to control the flow of loops.

22. What is method chaining in Java?

Method chaining is a technique where multiple methods are called on the same object in a single line. This is achieved by having each method return the current object. It's commonly used in builders and fluent APIs.

23. What are the types of polymorphism in Java?

Java supports two types of polymorphism: compile-time polymorphism (method overloading) and runtime polymorphism (method overriding). This allows one interface to be used for different underlying forms (data types).

24. What is the use of the 'super' keyword?

'super' refers to the immediate parent class of a class. It is used to access parent class methods and constructors, especially when they are overridden in the subclass.

25. What is the difference between 'public static void main' and other methods?

'public static void main(String[] args)' is the entry point of any standalone Java application. It must be static so the JVM can invoke it without creating an object, and 'void' means it returns no value.



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26. What are Java Packages?

Packages in Java are used to group related classes and interfaces together. They help organize code, avoid name conflicts, and control access using access modifiers. Examples include `java.util`, `java.io`, etc.

27. What is the difference between 'instanceof' and 'getClass()'?

'instanceof' checks if an object is an instance of a particular class or subclass. 'getClass()' returns the actual runtime class of the object and is used to compare exact class types.

28. What is a marker interface?

A marker interface in Java is an interface with no methods or fields. It is used to indicate or 'mark' a class for special behavior. Examples include `Serializable` and `Cloneable` interfaces.

29. What is the difference between 'throwable', 'error', and 'exception'?

'Throwable' is the superclass for all errors and exceptions. 'Error' represents serious issues that a program should not try to handle (e.g., `OutOfMemoryError`), while 'Exception' is used for recoverable problems.

30. What is a lambda expression?

Lambda expressions provide a clear and concise way to represent an anonymous function (a method without a name). They are primarily used to implement functional interfaces in Java, introduced in Java 8.